JOSHUA HEATH

Full-stack software developer | Final-year BSc Computer Science, University of Pretoria
Johannesburg, South Africa | +27 63 012 8337 | joshuaheath004@gmail.com
linkedin.com/in/joshua-heath-a910b5367 | github.com/Josh-Heath123

PROFILE

Final-year Computer Science student who builds and ships production software: the dashboard, anomaly-detection pipeline, MFA authentication and Terraform infrastructure of EquityLens, a live investment platform, and Culprit, a Go and Python tool that finds where a system contradicts itself. Open to graduate and internship software roles.

EDUCATION

University of Pretoria | BSc Computer Science 2023 – Dec 2026 (expected)
Relevant modules: Software Engineering, Compiler Construction, Computer Security, Database Systems, Software Modelling

St Dunstan's College | National Senior Certificate (Matric) 2022

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Go, Java, C++, SQL
Frameworks: React, FastAPI, SQLAlchemy, Alembic, Vite, Tailwind CSS
Cloud and DevOps: AWS (EC2, RDS, ECR, Cognito, Bedrock, SSM), Terraform, Docker, GitHub Actions, Git
Data and testing: PostgreSQL, Redis, MySQL, pytest, Vitest

SOFTWARE PROJECTS

EquityLens | Investment analytics for South African retail investors Apr 2026 – Present
COS 301 capstone, client: Amazon Web Services | Team of 4 | React, FastAPI, PostgreSQL, Terraform | equitylens.co.za

- Owned the main dashboard: portfolio value, a 0–10 health score built from concentration (HHI), largest position and breadth, and performance against a blended regional benchmark using time-weighted return and XIRR.
- Built the anomaly and news pipeline: an EWMA volatility model flags unusual moves ($|z| \geq 3$), an event study with beta splits each into market and company parts, and a nightly job ranks related news within a daily API budget.
- Moved authentication to AWS Cognito with TOTP multi-factor login and password reset, verified RS256 tokens on the backend against Cognito's public keys.
- Rebuilt production on AWS as Terraform modules and worked on the GitHub Actions pipeline that deploys through OIDC and SSM, migrates the database and swaps containers only after a health check passes.
- Largest contributor of four; CI blocks every merge on linting, type checks, security scans and 1,900+ automated tests at 88% backend coverage.

Culprit | Finds where a software system disagrees with itself Aug 2026 – Present
Personal project | Sole developer | Go, Python, TypeScript, React, PostgreSQL, Redis, Docker | Private repo, demo on request

- Wrote a Go CLI that compares what a repository says about itself (routes, OpenAPI specs, API calls, ORM models, migrations, env keys) and reports each contradiction, ranked by whether it breaks a request path.
- Runs in CI with no secrets: SARIF output annotates GitHub pull requests, and every run lists what it could not check, so an empty report is never mistaken for a clean one. A self-check of 983 facts gates its own pull requests.
- Built a Python incident-correlation service (FastAPI, arq, OpenTelemetry) that flags anomalies with EWMA z-scores and grounds root-cause suggestions in past postmortems via pgvector and the Claude API.

SPL Compiler | Compiler Construction (COS 341) | Java | Team of 5 2026

- Co-built a Java compiler for the course language SPL: lexer, parser, scope and type analysis, and C++ code generation with short-circuit Boolean jumps and function inlining.
- Verified it by differential testing: 900 random programs, built with g++ and gcc 13, matched an independent Python reference interpreter with zero disagreements.

CompareIt | Databases (COS 221) | PHP, JavaScript, MySQL | Team of 7 2025

- Built the login, registration and session flow, the products page and v2 of the API for a multi-retailer marketplace.

Plant Nursery Simulation | Software Modelling (COS 214) | C++17 | Team of 7 2025

- Built the Iterator pattern and its tests, the browser GUI and a cross-platform Makefile for a nursery simulation.

WORK EXPERIENCE

Part-Time Assistant | Family construction business 2023 – Present

- Measure sites, prepare material estimates and project costings, and handle client communication on site.